USN | | | | | | | | | | [22SEDE11]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

CLOUD COMPUTING AND VIRTUALIZATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**05x2=10**
1.	**a.**	List the usage of virtual box.	L1	CO1	**02**
	b.	What are the fundamental steps for setting up a cloud infrastructure?	L2	CO2	**02**
	c.	Distinguish between physical and virtual clusters.	L2	CO3	**02**
	d.	Criticize, how Azure Virtual Machines are used to run virtualized workloads in the cloud.	L2	C04	**02**
	e.	Compare the security and compliance features of AWS, Microsoft Azure, and Google Cloud Platform.	L2	C05	**02**
		PART - B			**6x5=30**
2.	**a.**	Compare public, private, and hybrid cloud models.	L2	CO1	**05**
		OR			
	b.	Explain in detail about the implementation level of virtualization.	L2	CO1	**05**
3.	**a.**	Illustrate the Infrastructure-as-a-Service, Platform as a service and Software as a service and explain.	L2	CO2	**05**
		OR			
	b.	Examine the use cases and advantages of object storage in cloud environments.	L3	CO2	**05**
4.	**a.**	How does block storage work? What are its typical applications in cloud computing?	L2	CO3	**05**
		OR			
	b.	Compare the different cloud load balancing services provided by AWS, Azure, and Google Cloud.	L2	CO3	**05**
5.	**a.**	Examine AWS and Microsoft Azure with cloud computing solutions for enterprises.	L3	CO4	**05**
		OR			
	b.	Outline the key features of Microsoft Azure and its primary services.	L2	CO4	**05**
6.	**a.**	Compare the use of Virtual Machines (VMs) and containers in cloud infrastructure management.	L3	CO5	**05**
		OR			
	b.	Interpret the Docker facilitate containerization in cloud platforms.	L4	CO5	**05**
7.	**a.**	Explain Kubernetes, and how is it used for orchestration in containerized environments.	L2	CO5	**05**

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		OR			
	b.	Explain AWS EC2 manage virtual machines in a cloud environment.	L2	CO5	**05**
		PART - C		**6x10=60**	
8.	**a.**	Explain the virtualization structure for **i.** Hypervisor and Xen Architecture. **ii.** Binary Translation with Full Virtualization. **iii.** Para-Virtualization with Compiler Support	L2	CO1	**10**
		OR			
	b.	Outline the advantages and limitations of each cloud service model (IaaS, PaaS, SaaS).	L2	CO1	**10**
9.	**a.**	Compare the following virtualization technologies: **i.** VMware **ii.** Hyper-V **iii.** KVM	L2	CO1	**10**
		OR			
	b.	Explain the different cloud deployment models. How do they differ from one another?	L2	CO2	**10**
10.	**a.**	Write short on note file storage. what scenarios is it most beneficial for cloud-based applications?	L2	CO3	**10**
		OR			
	b.	Determine the different storage tiers of Azure Blob Storage and their use cases.	L2	CO3	**10**
11.	**a.**	Examine the purpose and functionality of Virtual Private Networks (VPNs) in cloud environments.	L3	CO6	**10**
		OR			
	b.	Explain the Domain Name System (DNS) in cloud networking, and how does it facilitate resource access?	L2	CO6	**10**
12.	**a.**	Explain the Google Cloud Platform (GCP) for machine learning and big data applications.	L2	CO4	**10**
		OR			
	b.	Compare the security and compliance features of AWS, Microsoft Azure, and Google Cloud Platform.	L2	CO4	**10**
13.	**a.**	Explain the key virtualization and cloud management tools, such as vCenter, AWS EC2, and Azure Virtual Machines.	L2	CO5	**10**
		OR			
	b.	Examine, how Azure Virtual Machines are used to run virtualized workloads in the cloud.	L3	CO5	**10**